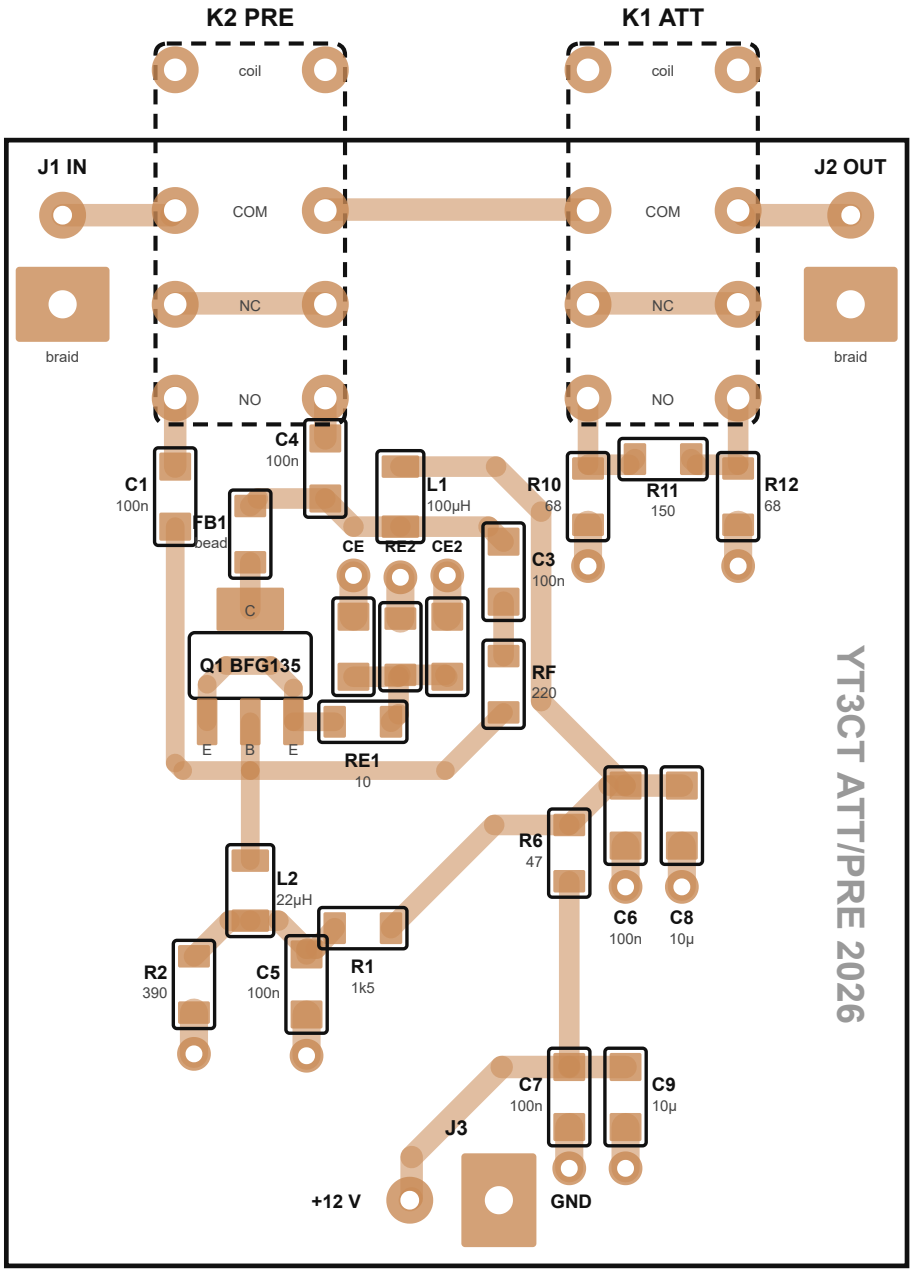
ATT / PRE board — component placement

Top view (component side, NOT mirrored), from the EasyEDA layout. Copper shown light, part outlines in black. Relays K1 / K2 go in from the bottom side.



Parts

Ref	Value	Note
Q1	BFG135	SOT-223, tab = collector (C), mark: pins E B E toward the bottom edge
RF	220 Ω	feedback, collector side → base
RE1 / RE2	10 Ω / 22 Ω	emitter
R1 / R2	1.5 k / 390 Ω	base bias
R6	47 Ω	+12 V feed to the preamp
R10, R12 / R11	68 Ω / 150 Ω	π-pad (16 dB)
C1, C3, C4, C5, C6, C7, CE	100 nF	1206
C8, C9, CE2	10 µF	1206, 25 V or more
L1 / L2	100 µH / 22 µH	1210; L1 rated ≥ 100 mA
FB1	ferrite bead	1206, right at the collector
K1 ATT, K2 PRE	Finder 30.22.7.012	from the BOTTOM side, soldered on top
J1 / J2	coax	core to the round pad, braid to the square pad
J3	wires	+12 V round pad, GND square pad

Order of work

1. After etching: beep every track against GND — no beep except the GND tracks.
2. Drill: relay pins and coax / J3 pads 1.2 mm, braid pads 1.5 mm, the 11 ground holes (small circles at the GND pads) 0.9 mm.
3. Bottom copper: countersink (ø 3 mm drill, by hand) every relay hole and the J1 / J2 / J3 **core** holes. Ground holes and braid holes stay connected.
4. Ground wires: bare wire through each of the 11 ground holes, solder top and bottom.
5. SMD flat parts first (resistors, capacitors), then L1, L2, FB1, then Q1 (tab soldered fully — heat).
6. Relays last, from the bottom, raised ~1 mm; solder the pins on top. Coil pins stay outside the board: flyback and sense diodes on them, wires to the rocker / Nano.
7. Coax: core to the round pad, braid flat onto the square pad (top), short pigtails.